

Globular Homotopy Theory of Spaces

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1 Introduction

Theorem 1: For any coherator, the Grothendieck–Quillen model structure on the category of models over the coherator exists. Furthermore, every topological realization of a coherator induces a Quillen equivalence between the category of spaces with the Quillen model structure and models of a coherator with the Grothendieck–Quillen model structure. Homotopies between topological realization give rise to natural weak equivalences between the corresponding Quillen adjunctions.

Theorem 2: Every cellular extension of cellular globular extensions that are equipped with standard model structures such that the right adjoint preserves cofibrations and acyclic cofibrations and unit applied to cell complexes is a weak equivalence induces a Quillen equivalence. If the source of the cellular extension receives a faithful globular functor from the globular completion with image in the subcategory of monomorphisms, the right adjoint preserves cofibrations and the unit is a weak equivalence under the additional hypothesis that the unit is a weak equivalence on globes. In particular, every cellular extension of models over a coherator with Grothendieck–Quillen model structures is a Quillen equivalence.

Theorem 3: Every Ara coherator is a coherator.

2 Cellular Globular Extensions

We name every maps in a cellular globular extension and compute the category explicitly.

3 Models over Cellular Globular Extensions

We name every element of the target of relative cell complexes.

We prove relative cell complexes are closed under the forgetful functor of a cellular extension of globular extension.

We prove that for a cellular extension of globular extensions such that the source receives a faithful globular functor from the globular completion whose image is contained in the subcategory of monomorphisms, the right adjoint of the induced adjunction between categories of models preserves cofibrations and the unit is a weak equivalence on cellular objects whenever it is a weak equivalence on representables.

We prove that every cellular extension of cellular globular extensions that are equipped with standard model structures such that the right adjoint preserves cofibrations and acyclic cofibrations and unit applied to cell complexes is a weak equivalence induces a Quillen equivalence.

4 Constructions in Coherators and Models

We construct various important morphisms in a admissibly filled cellular extension with chosen admissible fillings.

We construct the first frustum cylinder functor and prove its basic properties.

We construct the barycentric subdivision functor and prove its basic properties.

5 Frustum Cylinder Functor

We prove functoriality of the frustum cylinder in the comma category of relative cell complexes.

This is done in a way compatible with the next section.

6 Ara Coherators

We prove that every Ara coherator is a coherator.

7 Model Categories of Models over a Coherator

We prove that the Grothendieck–Quillen model structure exists on the category of models over a coherator.

We prove that a cellular extension of coherators induces a Quillen equivalence on Grothendieck–Quillen model structures.

8 Homotopy Hypothesis

We prove that a natural transformation of topological realization induces a natural weak equivalence between the associated Kan extensions to models.

We prove that a topological realization induces a Quillen equivalence between models over a coherator and spaces and homotopies between topological realization give rise to natural weak equivalences between the corresponding Quillen adjunctions.